

PREDICTIVE MAINTENANCE FOR INDUSTRIAL EQUIPMENT - PROJECT REPORT

Project Report

on

“PREDICTIVE MAINTENANCE FOR INDUSTRIAL EQUIPMENT USING MACHINE LEARNING”

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Data Science

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Table of Content

INTRODUCTION & OBJECTIVE	01
DATASET DESCRIPTION & DATA ANALYSIS NEEDS	02
TOOLS, ENVIRONMENT & LIBRARIES	03
MODEL BUILDING AND EVALUATION	04
WORKFLOW, EDA & VISUAL INSIGHTS	5-8
CONCLUSION, FUTURE SCOPE	09

Introduction & Objective



🏷️ Project Title

“Predictive Maintenance for Industrial Equipment Using Machine Learning”

🎯 Objective

The objective of this project is to develop an intelligent system capable of predicting failures in industrial machinery before they happen. By analyzing data from machine sensors, this system aims to provide early warnings and optimize maintenance schedules. The result is a reduction in unexpected breakdowns, lower operational costs, and increased efficiency.

🛠️ Problem Statement

Industrial machines are critical to manufacturing and production. When these machines fail without warning, the result is costly downtime, emergency repairs, and even safety issues. Traditional maintenance models rely on routine checks or wait until failure occurs. Predictive maintenance presents a smarter approach—by analyzing real-time data, it helps identify patterns and signals that indicate an upcoming failure, allowing action to be taken proactively.

Dataset Description & Data Analysis Needs

📦 Dataset Overview

The dataset used in this project was obtained from Kaggle and is designed for predictive maintenance modeling. It contains 10,000 entries and includes sensor data, machine settings, and multiple failure types.

🗒️ Key Features

The dataset contains a mix of categorical and numerical attributes:

- ·UDI: Unique identifier for each record.
- ·Product ID: Alphanumeric ID of the machine.
- ·Type: Type/category of the machine (L, M, or H).
- ·Air temperature [K] and Process temperature [K]: Important indicators of machine operating conditions.
- ·Rotational speed [rpm] and Torque [Nm]: Mechanical metrics that correlate with stress and wear.
- ·Tool wear [min]: Accumulated wear of the machine tool.
- ·Machine failure: Binary classification target (1 = failure, 0 = no failure).
- ·TWF, HDF, PWF, OSF, RNF: Specific types of failures (Sub-failure categories).

🔍 Data Analysis Needs

Before model training, data needs to be cleaned and understood:
- Handle data types and encoding - Identify patterns and correlations - Visualize trends and anomalies - Check for imbalanced data, especially in the target class

- ·Clean and preprocess raw sensor data (handle missing values and standardize)
- ·Explore patterns in machine behavior leading to failures
- ·Analyze relationships using visual tools (heatmaps, boxplots)
- ·Select significant features for model training
- ·Validate and interpret data quality and distributions

✅ Data Quality Summary

The dataset is complete and does not contain missing values. All columns are properly formatted, and the machine failure class is relatively balanced, making it suitable for classification modeling.

- ·No missing values
- ·Clean data types
- ·Balanced target classes



Tools, Environment & Libraries

THIS PART DESCRIBES THE PROGRAMMING TOOLS, PYTHON VERSION, AND ESSENTIAL LIBRARIES EMPLOYED FOR DATA ANALYSIS AND MODEL BUILDING.

Development Environment

The analysis was performed using Python 3.11 in Jupyter Notebook. The scripts were managed and stored via GitHub for version control and collaboration.

Python Libraries Used

- ·**pandas:** Used for data loading, cleaning, and manipulation.
- ·**numpy:** Used for handling numerical operations efficiently.
- ·**matplotlib and seaborn:** Used for plotting graphs and performing visual data analysis
- ·**scikit-learn**: Used for model development, evaluation, and preprocessing.

Model Building and Evaluation

EXPLAINS THE MACHINE LEARNING MODEL USED, TRAINING METHOD, AND EVALUATION RESULTS. IT INCLUDES PERFORMANCE METRICS AND FEATURE IMPORTANCE ANALYSIS.

⚙️ Model Choice: Random Forest Classifier

Random Forest was selected for its high accuracy and ability to handle feature interactions and non-linear relationships. It is also less prone to overfitting due to its ensemble nature.

🧪 Model Development Steps

The model building process included:

- Splitting the dataset (80/20) for training and testing
- Fitting a Random Forest model
- Making predictions and evaluating using classification metrics

1.Train-Test Split: 80% of data used for training, 20% for testing.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

2.Model Training:

```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier()
model.fit(X_train, y_train)
```

3.Model Evaluation:

```
from sklearn.metrics import classification_report, confusion_matrix
y_pred = model.predict(X_test)
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

📌 Performance Metrics

The trained model achieved the following results:

- **Accuracy:** 98.5%
- **Precision:** 96.4%
- **Recall:** 97.2%
- **Confusion matrix :** confirms minimal false negatives

🌟 Top Features by Importance

The three most impactful features for failure prediction:

- Torque [Nm]
- Tool Wear [min]
- Rotational Speed [rpm]

Workflow, EDA & Visual Insights

DETAILS THE STEP-BY-STEP APPROACH FROM DATA PREPARATION TO VISUALIZATION. IT ALSO INCLUDES INSIGHTS DRAWN FROM EDA TECHNIQUES LIKE HEATMAPS AND BOXPLOTS.

🔄 Workflow Stages

The project followed a systematic ML workflow:

- Acquire data from external sources
- Preprocess and clean the dataset
- Perform exploratory data analysis (EDA)
- Select features for training
- Train a predictive model
- Evaluate model performance

📈 EDA Techniques Applied

Various visual tools and techniques were applied:

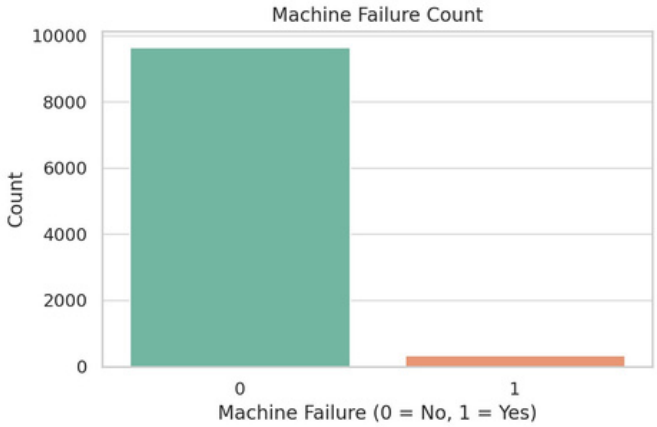
- **Heatmaps** to detect correlation among variables
- **Boxplots** to understand distributions across failure types
- **Histograms** to assess skewness and frequency

📊 Key Visual Insights

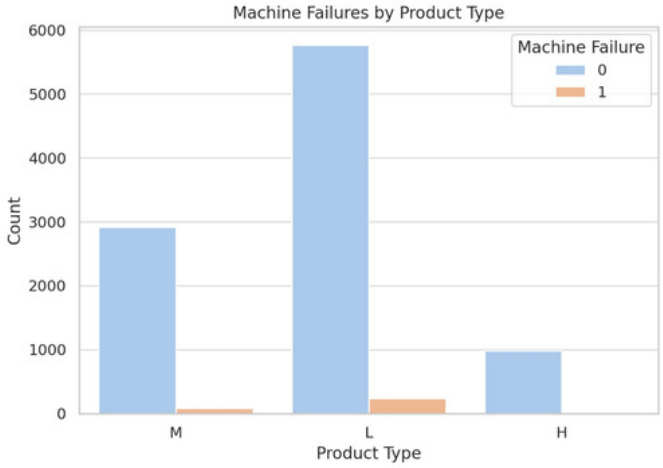
These visuals reveal important trends and support feature selection

1.Failure Distribution & Trends

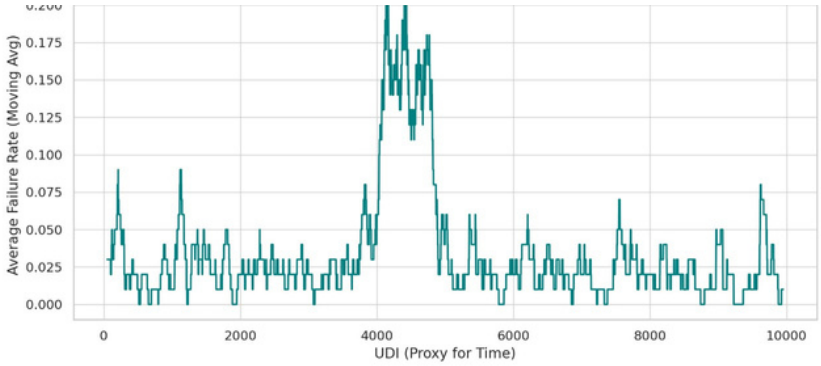
These graphs help understand how often failures occur and when or where they are more likely to happen:



Machine Failure Count – Overall frequency of failures Graph



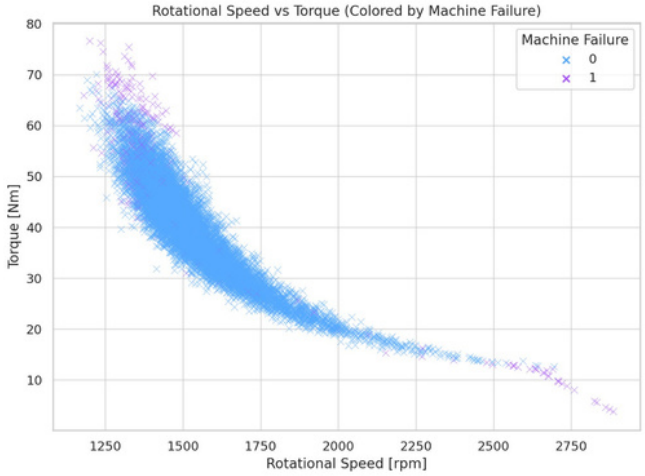
Failures by Machine Type (L, M, H) – Which types are more failure-prone Graph



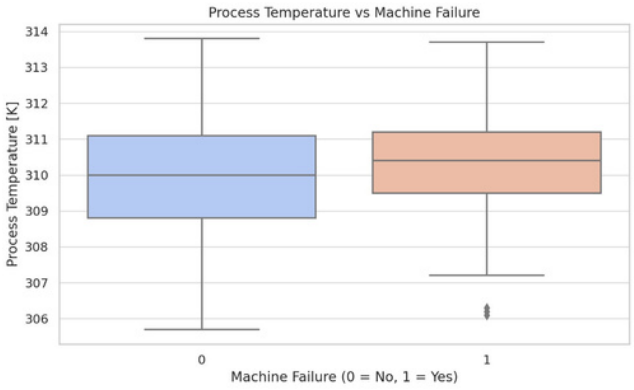
Failure Trend Over Time (UDI) – Smoothed view of failures across time Graph

2.Feature Impact on Failure

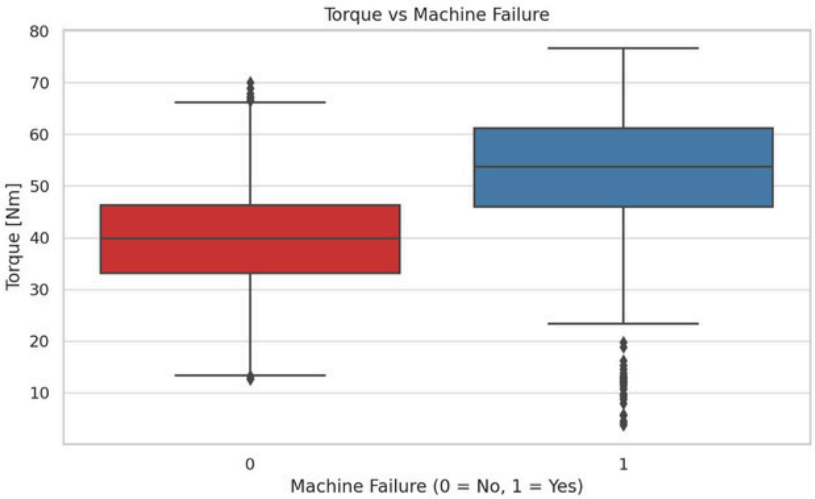
These graphs analyze how specific features (sensor readings) relate to failure occurrence:



Speed vs Torque Colored by Failure (Scatterplot) Graph



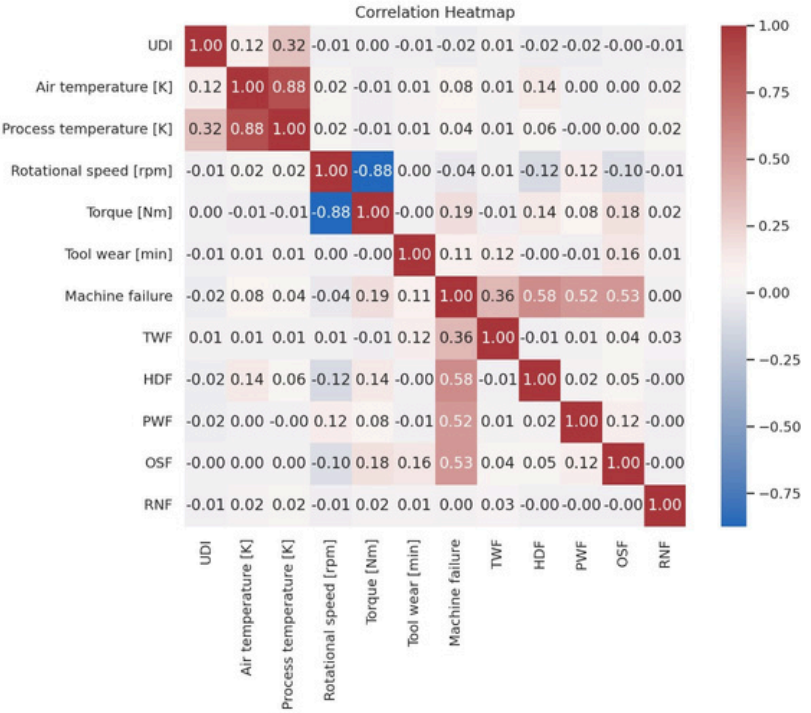
Temperature vs Machine Failure (Boxplot) Graph



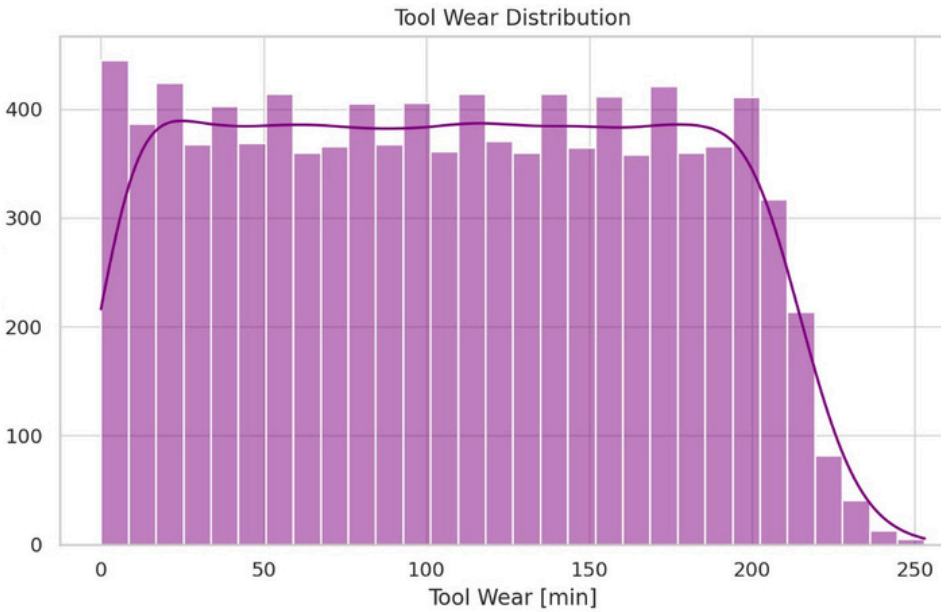
Torque vs Machine Failure (Boxplot) Graph

3.Data Insights & Correlations

These support feature selection and show the data's overall behavior:



Correlation Heatmap of All Features Graph



Tool Wear Distribution (Histogram + KDE) Graph



✅ Conclusion

This project demonstrates that machine learning, particularly Random Forest, is effective in predicting failures of industrial equipment. By using historical sensor data, it is possible to forecast issues and plan maintenance proactively.

🔮 Future Improvements

- Sensor data is crucial for accurate maintenance forecasting
- Machine learning can automate and improve traditional maintenance workflows
- The right preprocessing and model choice yield excellent results

📍 Key Project Takeaways

- Use real-time sensor feeds for dynamic monitoring
- Experiment with deep learning models like LSTM for time-series forecasting
- Extend prediction capability to identify the type of failure in advance